

CPSC-406 Report

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Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$. Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$ (columns: Accepted by $\mathcal{A}_1$? and Accepted by $\mathcal{A}_2$?).

Exercise 1.2

Describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain aba

3. All the words that contain an odd number of a 's and an odd number of b 's
4. All the words that contain an even number of a 's and an odd number of b 's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain bbb

What do you notice when comparing the various automata?

Answers:

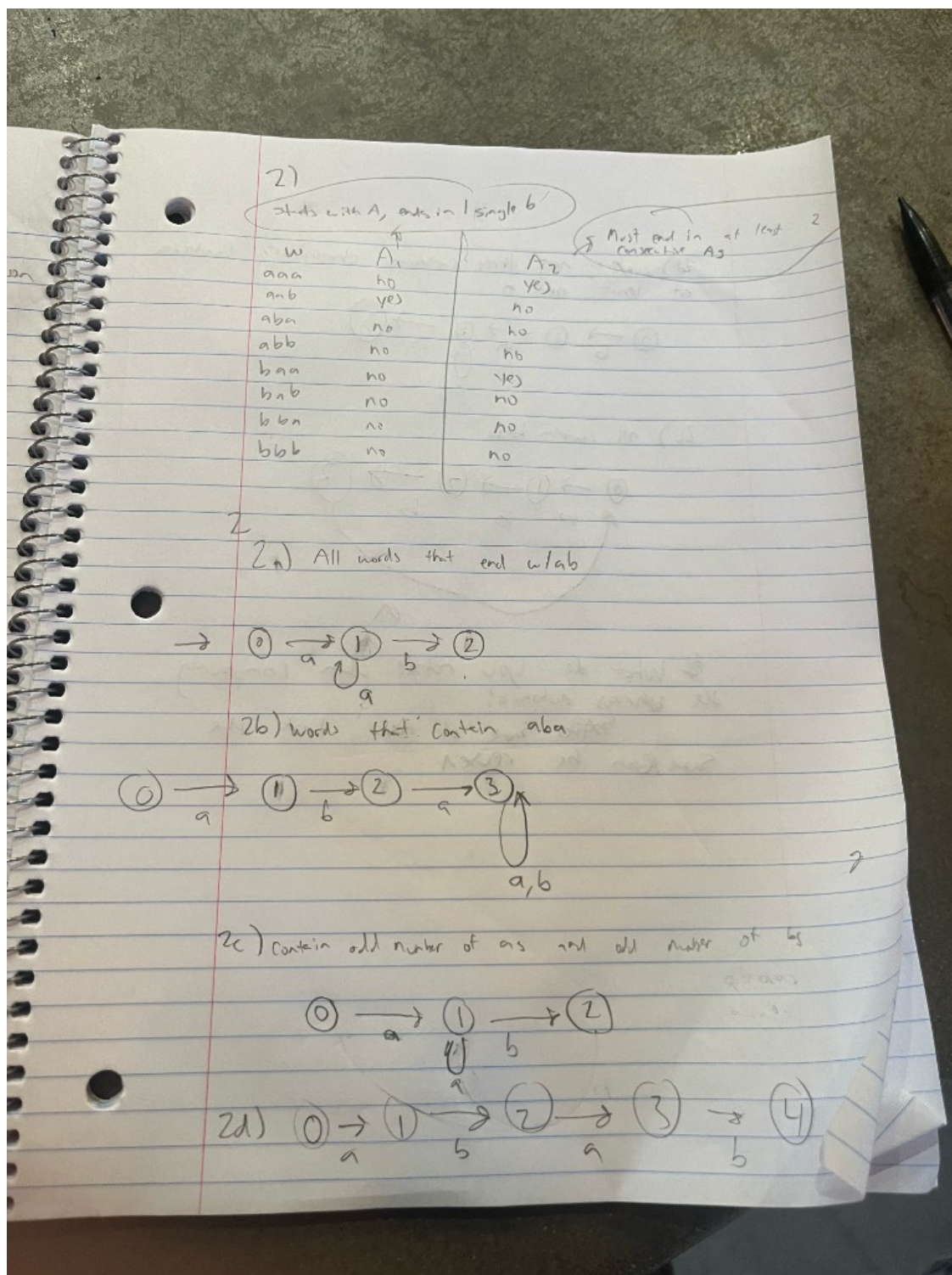
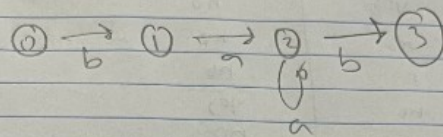
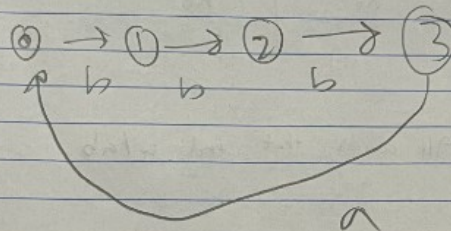


Figure 1: Week 1 answers (table, language conditions, DFA designs 2a-2d).

2d) any three consecutive characters contain at least one a



2c) all contain bbb



What do you notice when comparing the two automata?
 Some of them look the same/can be reused

Figure 2: Week 1 answers (DFA designs 2c, 2d; comparison of automata).

Question

How might you combine the languages $L(\mathcal{A}_1)$ and $L(\mathcal{A}_2)$ from Exercise 1 using set operations (union, intersection, complement)? Could you represent the logic with boolean logic exclusively?

2.2 Week 2: Operations on Automata

Product automata

Exercise 1 (Product automata)

Consider the following automata $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$.

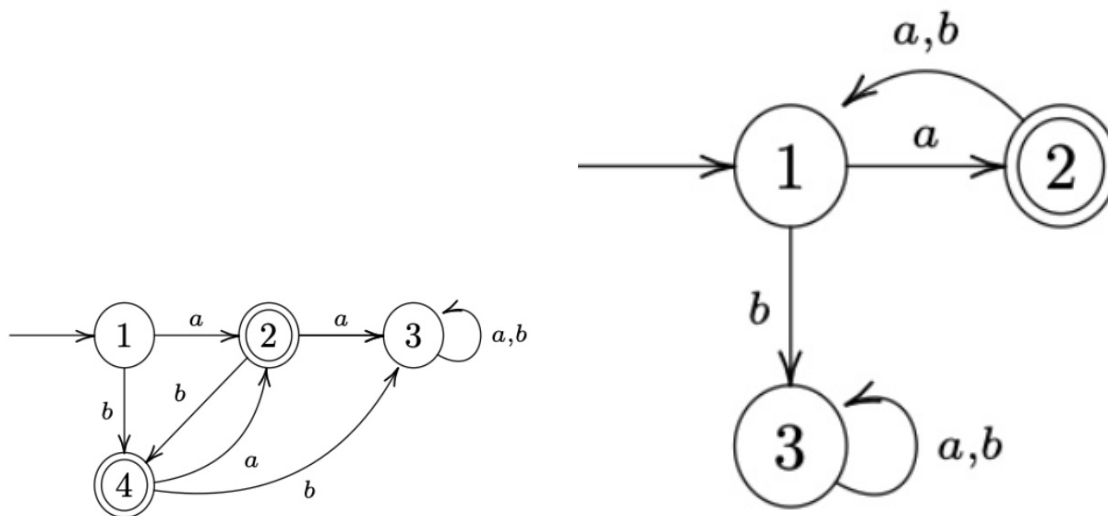


Figure 3: $\mathcal{A}^{(1)}$ (left) and $\mathcal{A}^{(2)}$ (right).

1. Describe $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$.

Answer:

- $L(\mathcal{A}^{(1)})$: Words include $a, b, ba, bab, baba, ab, aba, abab$, etc. I can see that each character must alternate based on its successive character.
- $L(\mathcal{A}^{(2)})$: Words include a, axa (where x doesn't matter). I see the word must start with a , and must add to its word length in increments of 2, every other character being a .

2. Construct the intersection automaton $\mathcal{A}$ for $\mathcal{A}^{(1)}$ and $\mathcal{A}^{(2)}$ by drawing its transition diagram (omit unreachable states).

Answer: Below is the intersection automaton diagram.

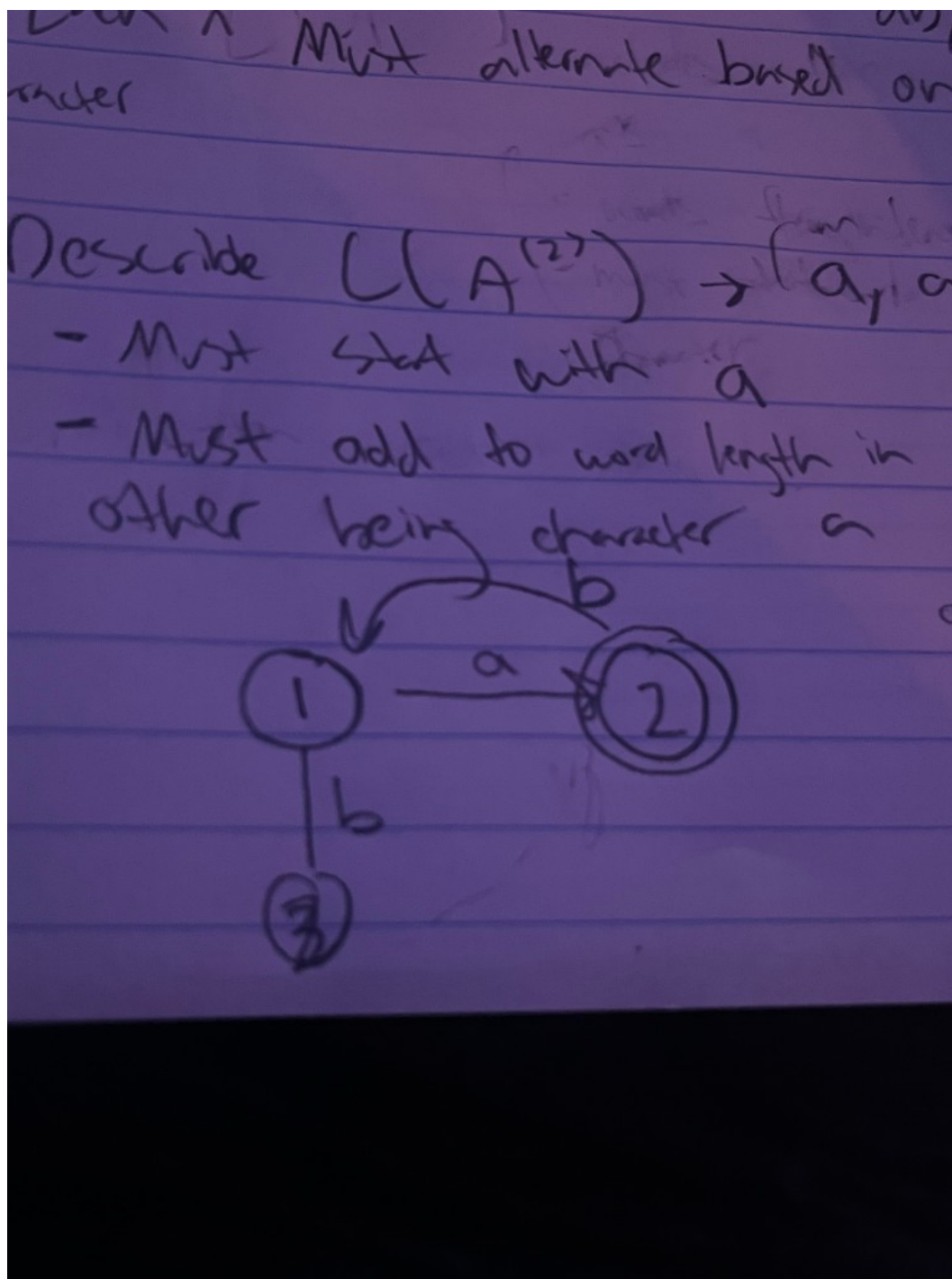


Figure 4: Intersection automaton $\mathcal{A}$ (hand-drawn diagram).

3. **Argue that** $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$.

Answer: Since $L(\mathcal{A}^{(1)})$ includes words such as a , b , ba , bab , $baba$ and ab , aba , $abab$, and $L(\mathcal{A}^{(2)})$ includes words that (1) must start with a —so we can disregard the words that start with b in $L(\mathcal{A}^{(1)})$,

leaving us with a, ab, aba , etc.—and (2) the requirement for $L(\mathcal{A}^{(2)})$ is to be in increments of two, we are left with patterns: $a, aba, ababa$, etc. Thus we obtain the automaton of both intersection.

4. **How can you change $\mathcal{A}$ to get an automaton $\mathcal{A}'$ with $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$?**

Answer: We can allow for both word combinations to be accepting and add more state transitions.

Exercise 2 (More automata)

Consider the following automata $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$.

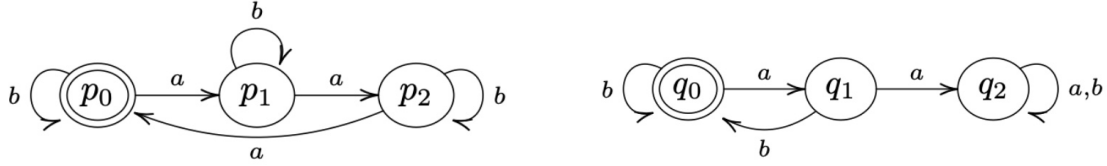


Figure 5: $\mathcal{B}^{(1)}$ (left) and $\mathcal{B}^{(2)}$ (right).

(Recall: the complement of a language $L \subseteq \Sigma^*$ is $\bar{L} := \Sigma^* \setminus L$.)

1. **Describe $L(\mathcal{B}^{(1)})$ and $L(\mathcal{B}^{(2)})$.**

Answer:

Figure 6: $L(\mathcal{B}^{(1)})$ rules: include 3 a 's at a time, any amount of b 's. $L(\mathcal{B}^{(2)})$ rules: b must follow an a , any amount of b 's. DFA diagram for language 2.

2. **Construct the intersection automaton $\mathcal{B}$ for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ (omit unreachable states).**

Answer: Below is the intersection automaton diagram.

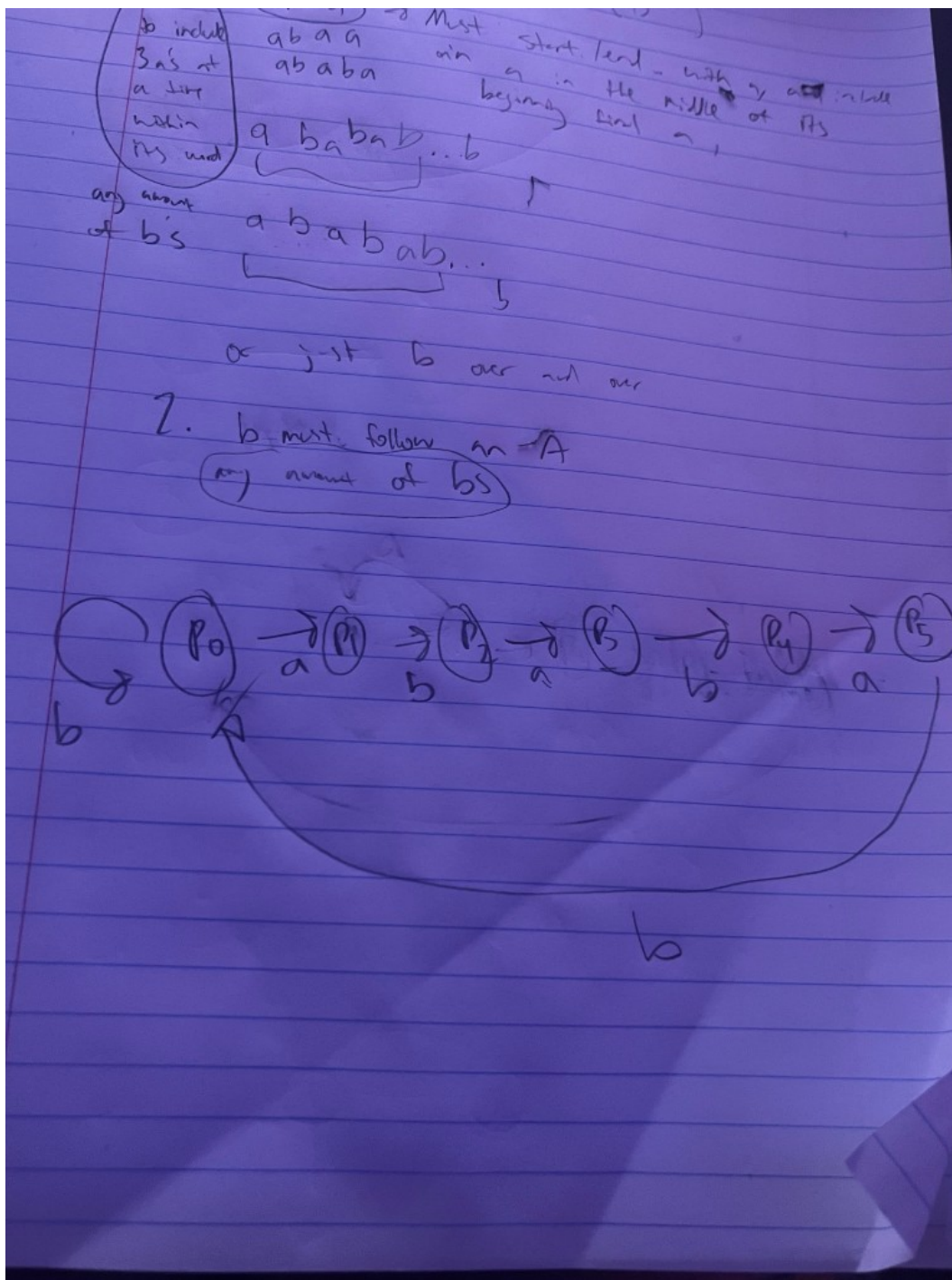


Figure 7: $L(\mathcal{B}^{(1)})$ rules: include 3 a's at a time, any amount of b's. $L(\mathcal{B}^{(2)})$ rules: b must follow an a, any amount of b's. DFA diagram for language 2.

3. Argue that $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$.

Answer: Both automata have the rule that any amount of b's is allowed. But we must combine the

rules regarding a 's. $L(\mathcal{B}^{(1)})$ requires 3 a 's to be included at a time per word, while $\mathcal{B}^{(2)}$ requires the rule that a must be followed by a b . This gives us the conclusion that words that do include an a must come in pairs of 3 a 's with b following the last a .

4. **How can you change $\mathcal{B}$ to get an automaton $\mathcal{B}'$ with $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$?**

Answer: We must flip the accepting states of automaton 1 to include: no amount of b 's AND words that include 3 a 's at a time only.

Exercise 3

Let $\mathcal{A}$ be a DFA and q a state of $\mathcal{A}$ such that from q , every input symbol a leaves us in q (i.e. $\delta(q, a) = q$ for all a). Show by induction on the length of the input that for any input string w , we still end up in q (i.e. $\delta^*(q, w) = q$).

Answer: We show it by induction on the length of w . (Here $\delta^*(q, w)$ is just the state we end up in after starting at q and reading the whole string w .)

When the input has length 0, the string is empty. So we don't read anything and we stay at q . So $\delta^*(q, w) = q$ in that case.

Now suppose we've already shown that for every string of length n , we end up back at q . Take a string w of length $n + 1$. We can write it as $w = va$ (some string v of length n plus one symbol a). After reading v we're in $\delta^*(q, v)$, and by our assumption that's q . Then we read a , and we're in $\delta(q, a)$, which is q by the given rule. So after reading w we're still in q , i.e. $\delta^*(q, w) = q$.

So by induction, for all input strings w we have $\delta^*(q, w) = q$.

Question

If you build the product automaton for $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ but change the accepting set to get the *union* $L(\mathcal{B}^{(1)}) \cup L(\mathcal{B}^{(2)})$ instead of the intersection, how many states must be accepting?

2.3 Week 3

Homework 1 (NFA)

1. Describe the language accepted by $\mathcal{A}$ using a regular expression.
2. Specify the NFA $\mathcal{A}$ in the form $(Q, \Sigma, \delta, q_0, F)$.
3. Find all paths in $\mathcal{A}$ for the word $w = abab$; represent in a diagram.

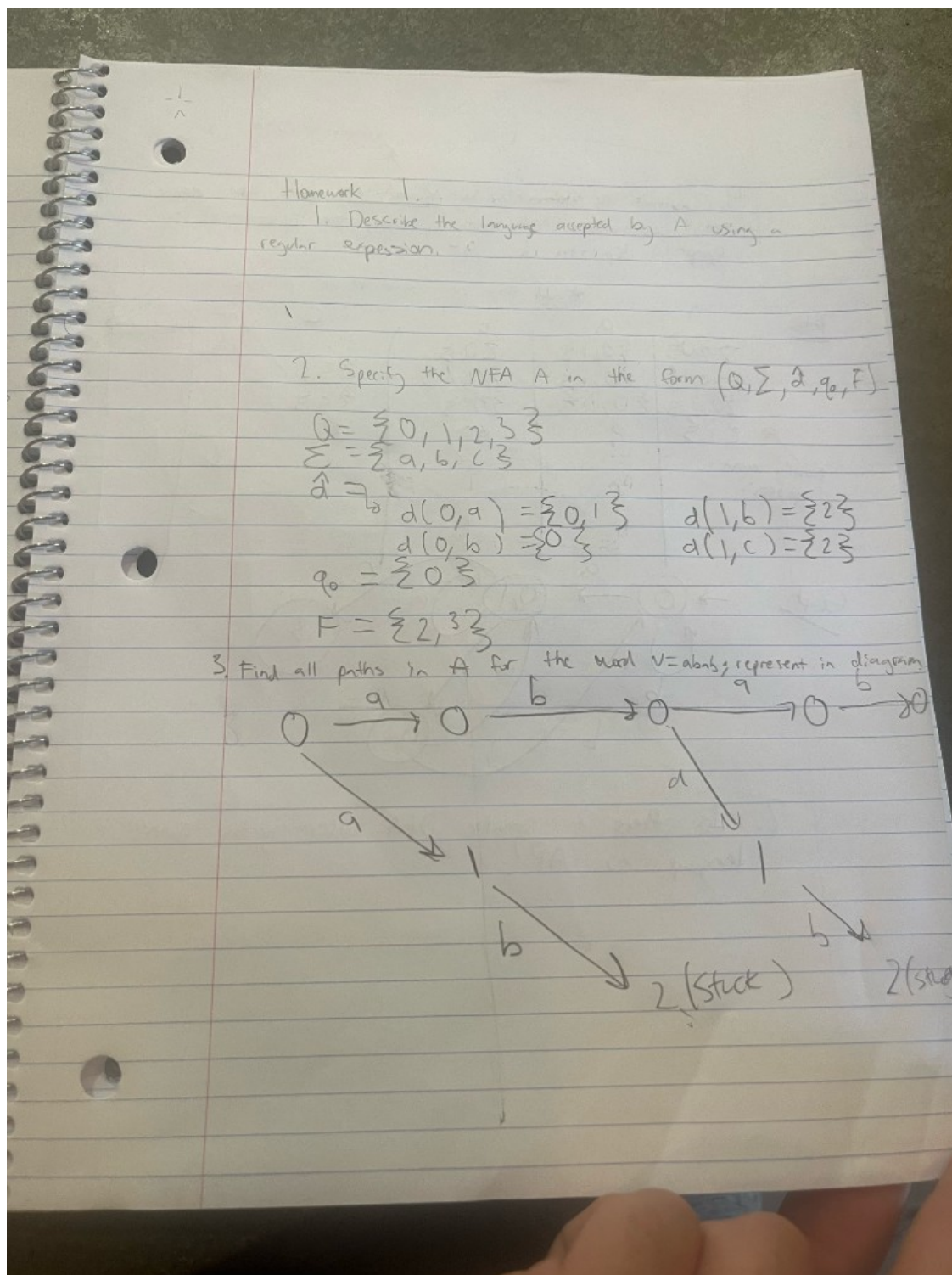


Figure 8: Homework 1: NFA specification $(Q = \{0, 1, 2, 3\}, \Sigma = \{a, b, c\}, q_0 = 0, F = \{2, 3\})$ and all paths for $w = abab$.

Homework 2 (Determinization)

Construct the determinization $\mathcal{A}^D$ of $\mathcal{A}$ using the power set construction. Create both a transition table and a graphical depiction of $\mathcal{A}^D$.

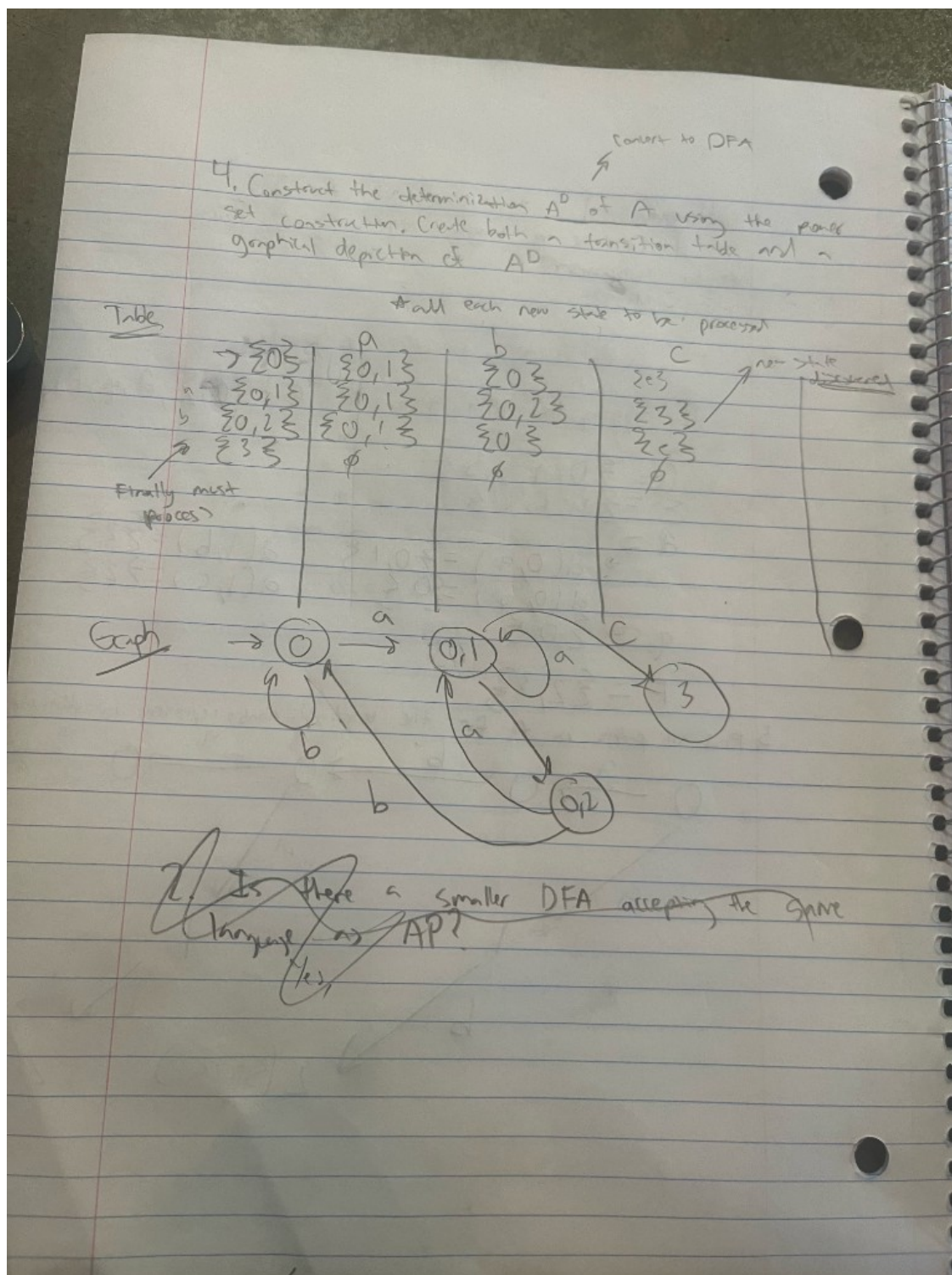


Figure 9: Power set construction: transition table and graph of A^D .

Homework 3 (DFA and minimization)

Determine $\mathcal{A}^D$ of $\mathcal{A}$ using the power set construction. Create both a transition table and a graphical depiction of $\mathcal{A}^D$. Then: can there be a smaller DFA accepting the same language? (Yes: the last two rows in the table are identical, so we can minimize by one state.)

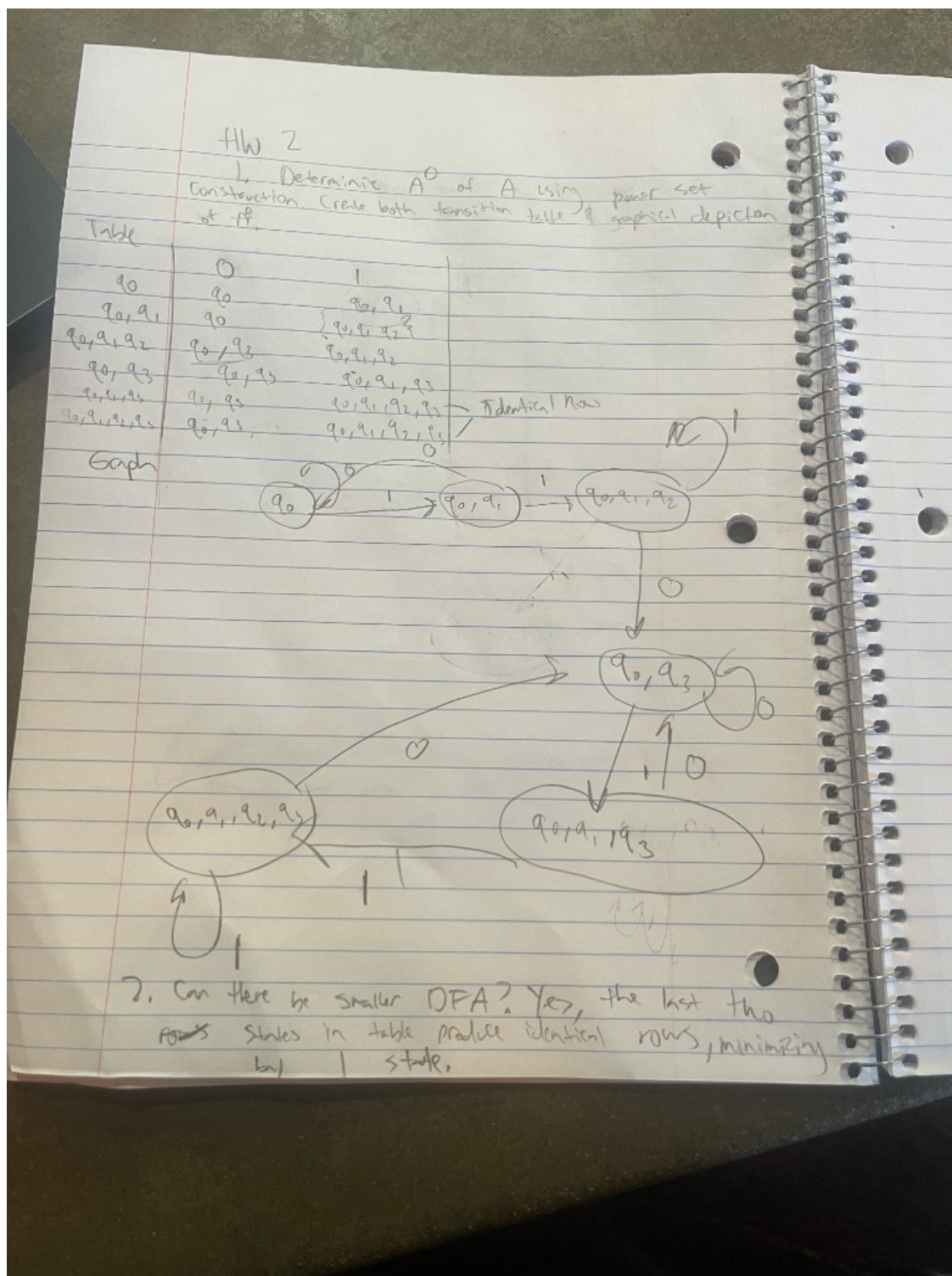


Figure 10: Power set construction with alphabet $\{0,1\}$; table and graph of A^D ; minimization by merging identical rows.

Question

After building $\mathcal{A}^D$ from an NFA $\mathcal{A}$ using the power set construction, when can we get a smaller DFA that still accepts the same language?

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, [Title](#), Publisher, Year.